YOUTUBE TRENDING VIDEOS ANALYTICS

Project Report
Tools Used: Python, SQL and Tableau

1. Introduction

YouTube is a popular platform where videos become trending based on viewers' interests and engagement. This project analyzes trending YouTube videos across different countries to understand their performance. The main objective is to explore video views, likes, categories, title sentiment, and trending duration using data analytics tools.

2. Abstract

This project focuses on analyzing YouTube trending video data using Python, SQL, and Tableau. Python was used for data cleaning, data analysis, and sentiment classification of video titles. SQL queries helped explore popular videos and summarize performance. Tableau was used to create charts and KPI cards to present the findings. The final dashboard provides a visual overview of country-wise performance, category-wise views, title sentiment, and trending duration.

3. Tools Used

- Python (Google Colab): Data cleaning and analysis.
- Pandas and NumPy: Data manipulation and calculations.
- VADER Sentiment Analysis: Approximate sentiment classification of video titles.
- SQL (SQLite): Querying and summarizing data.
- Tableau: Creating visualizations, KPI cards, and the dashboard.
- GitHub: Sharing project files and code.

4. Steps Involved in Building the Project

1. Collected country-wise YouTube trending video CSV files.
2. Loaded the data into Google Colab and inspected its columns and quality.

3. Cleaned the data by handling encoding issues, malformed rows, and invalid category values.
4. Used Python to analyze country performance, video categories, title sentiment, and trending duration.
5. Prepared summary CSV files for visualization.
6. Used SQL queries to explore popular videos and summarize performance.
7. Imported the summary files into Tableau and created charts and KPI cards.
8. Combined the visualizations into a dashboard and saved the Tableau Packaged Workbook.

5. Conclusion

This project provided practical experience in data cleaning, exploratory analysis, SQL querying, and dashboard creation. The analysis highlighted differences in video performance across countries and categories, the distribution of title sentiment, and variations in trending duration. Some source data required cleaning, and VADER sentiment results should be interpreted carefully because the dataset contains titles in multiple languages. Overall, the project demonstrates how data analytics tools can be used to explore and present YouTube trending video patterns.